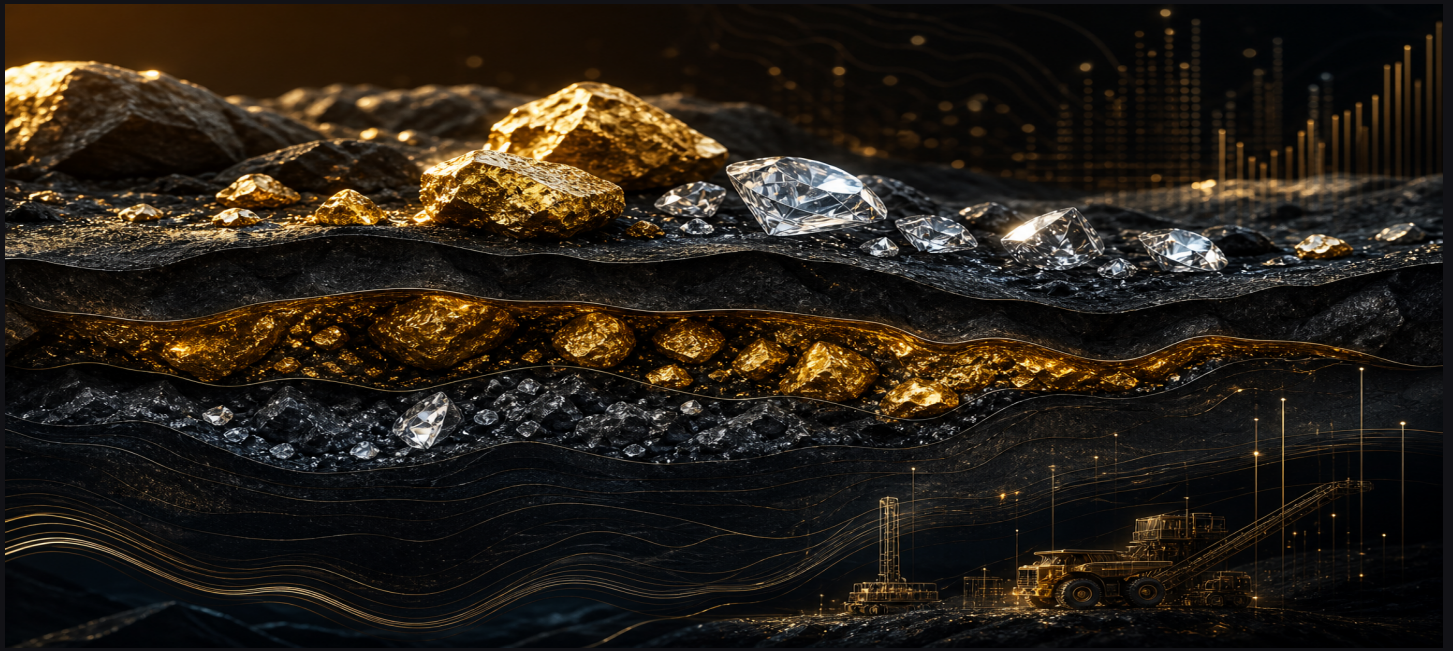
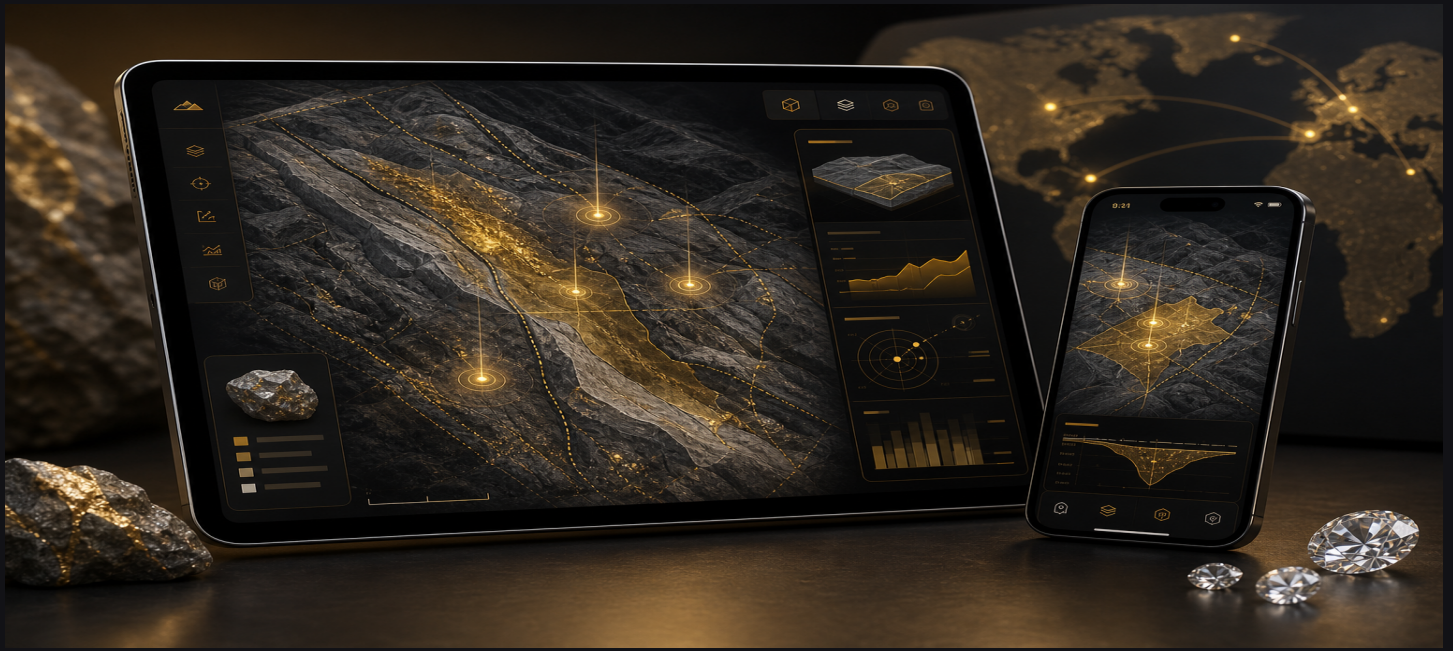




1. Executive Summary

WCS Mining AI is a mobile-first exploration intelligence platform for World Class Scholars. It helps teams decide where to explore, why targets rank highly, and what evidence supports each recommendation. The product combines public Australian geoscience, tenement data, drilling, satellite layers, and tenant-private imports into explainable prospectivity scoring - delivered through a premium SwiftUI experience with sparkling gold and diamond branding.

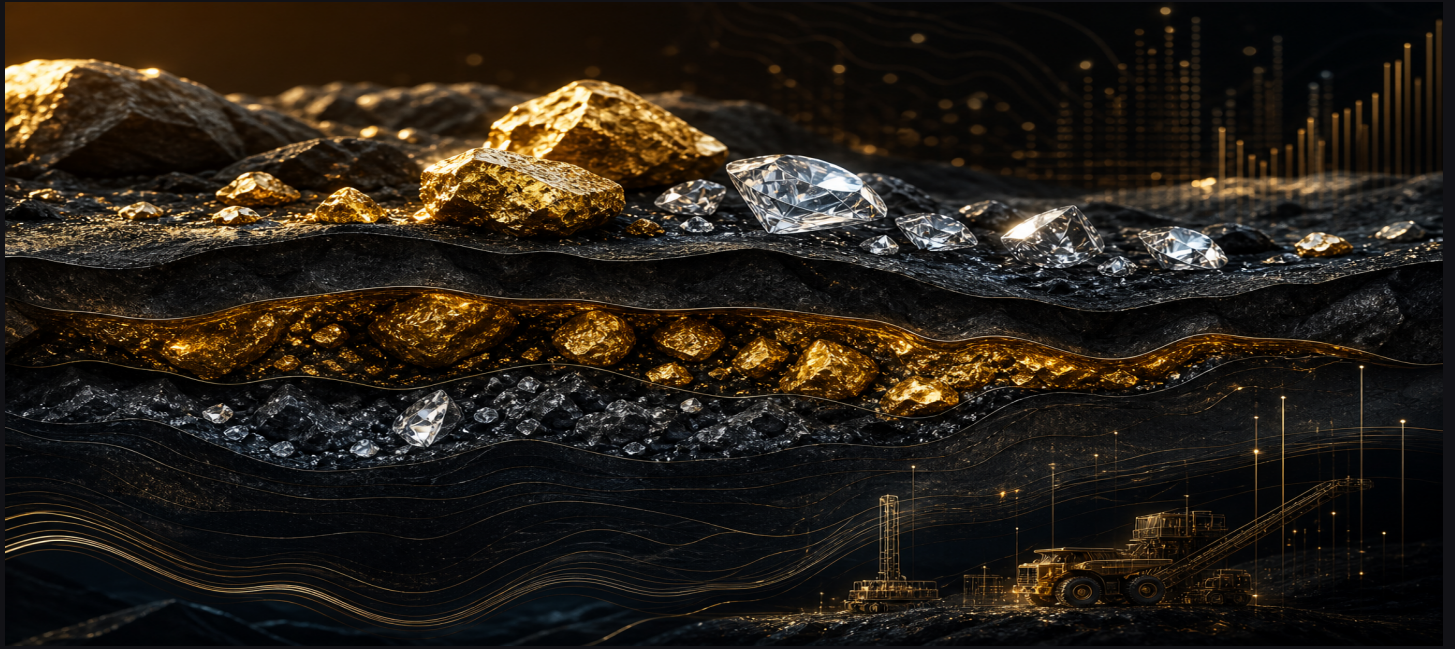
Sparkling gold and diamonds brand | Apple production assets included



2. Problem & Market Opportunity

Junior and mid-tier explorers lack affordable, field-ready AI that geologists can trust. Desktop suites are powerful but slow in the field; black-box heatmaps fail board and field scrutiny. Australia offers a strong wedge: rich public data, Digital Earth Australia rasters, and active gold/critical-minerals campaigns.

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3. Map Explorer Module

The map explorer unifies tenements, geology, occurrences, drillholes, raster overlays, and AI-ranked targets. Users toggle layers, filter by commodity, download offline packs, and open target evidence from the map. MapKit provides the base map; custom vector and raster overlays connect to PostGIS-backed project layers.

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4. Target Ranking Module

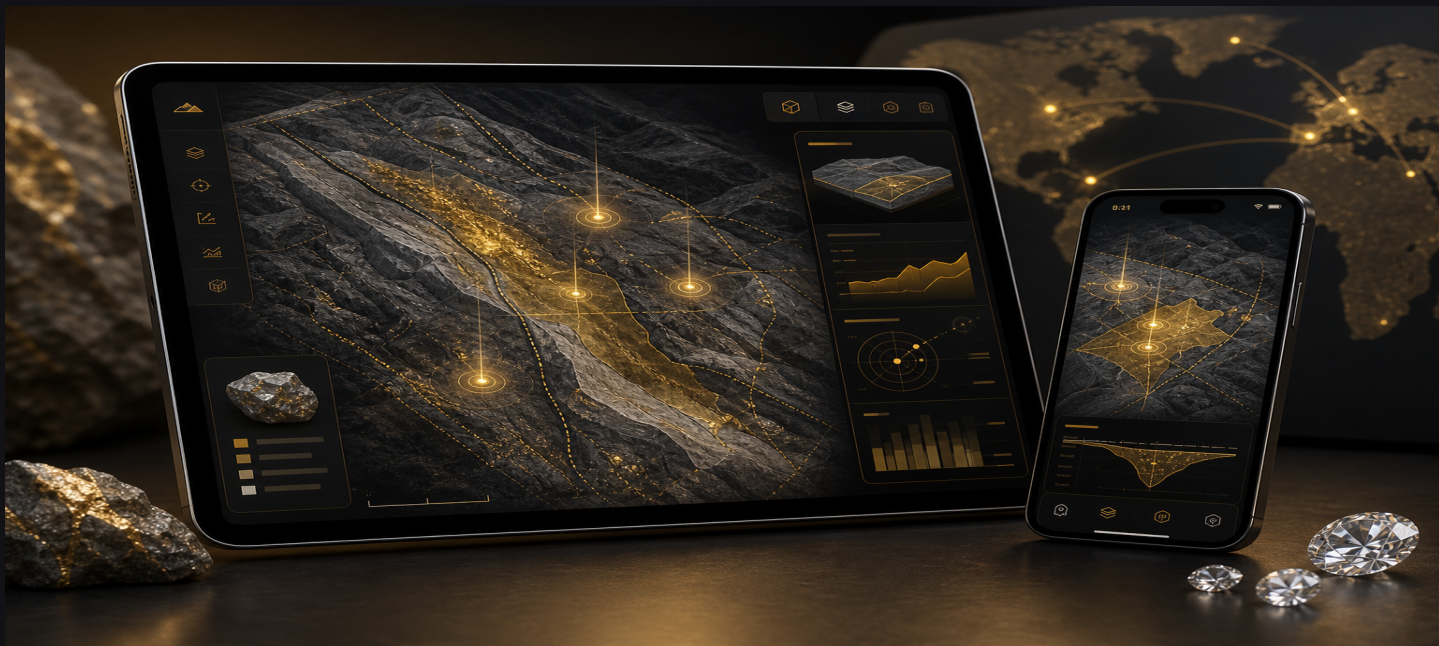
Targets are ranked cells or merged polygons with prospectivity score, confidence band, commodity, depth hypothesis, model version, and decision status. Scoring uses a gridded feature architecture (data-cube pattern) with batch regional runs and on-demand rescoring after private imports.



5. Explainability Module

Each target exposes top-five feature drivers, analog deposit similarity, data completeness, last rescore date, and a geological narrative. Evidence is tagged Public, Private, or Blended so teams know which inputs are company-confidential. Gradient-boosted trees and random forests prioritize interpretability over opaque deep models in early releases.

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6. Data Room & Private Fusion

The data room ingests assay CSV, collar surveys, GeoJSON structures, and legacy exports. Imports run as asynchronous jobs with provenance and validation reports. Tenant isolation via Supabase RLS ensures private drilling and assays never leak across organizations.

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7. Field Operations

Geologists download targets, review evidence offline, capture observations, annotate outcrops, and attach photos. Core Location anchors observations; offline UUIDs enable conflict-safe sync. Background upload resumes when connectivity returns - critical for remote Australian exploration.

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8. Architecture & Security

SwiftUI client; Supabase Postgres + PostGIS; Supabase Storage; Python ML pipelines; edge/Nest orchestration. API surface: projects, map-layers, targets, observations, imports, rescore, model summary. Roles: geologist, exploration manager, data manager, executive viewer - with audited imports and rescoring.

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9. Roadmap & Commercial Model

Phase 1: Australia gold MVP with public layers and map/target viewer. Phase 2: private imports, explainability, field photos. Phase 3: multi-commodity, uncertainty, analyst dashboards. Phase 4: enterprise approvals and GIS integrations. SaaS per organization with private-data and rescoring upsell.

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10. Investment Highlights

WCS differentiation: explainable targeting, iPad-first field workflow, and education/simulation mode for training partnerships. Premium brand identity (sparkling gold & diamonds) supports App Store and investor materials. CI/CD on GitHub ensures repeatable iOS builds. MVP acceptance criteria are defined and tracked in the canonical product spec.

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